



MINING DEALER BEST PRACTICE

3600 / C280 Cam Follower Holding Tools

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

When completing any work on 3600 / C280 engine camshafts, cam bearings, or timing gears you need to traditionally take the time to remove the rocker gear, push rods and cam followers.

2.0 Best Practice Description

This Best Practice outlines the use of a hand-crafted tool that eliminates the need to remove all rocker gear, push rods and followers when completing work on 3600 / C280 camshafts, cam bearings and timing gears as stated in the D&A manual.

By connecting a key ring (or some other durable, highly visible tag) to a standard split pin you have a tool that can save lots of disassembly and assembly time. The 3600 Disassembly & Assembly manual states that you must first remove the valve mechanism and valve lifter groups before camshaft work. However by using this tool and procedure you only need to loosen off the valve mechanism on the cam that you are working on. By lifting and inserting the tool into the cam follower oil hole it holds up the weight of the valve mechanism and valve lifter groups.



Cam Follower Holding Tool



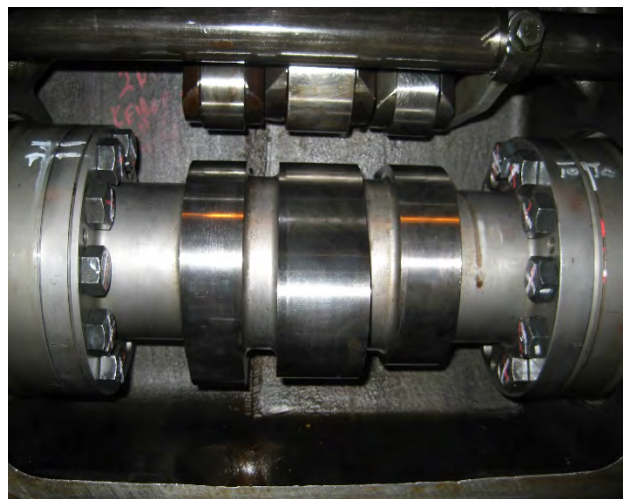
Tools x3 installed into Cam Followers

The photo above on the left shows a simple tool that can be manufactured. The photo on the right shows three (3) of the tools holding up the followers on one cylinder. Note in these photos the rocker gear and push rods have been removed (for other reasons) but this is not necessary when using this tool during camshaft work.

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Cam Follower Holding Tools being used



Cam Followers being held up clear of Camshaft

The split pin does all the work but it's critical to have a highly visible tag attached to avoid the tool being overlooked and missed on reassembly. The tags also stop the pins falling and getting lost in the cam area or sump if they are accidentally dropped. In the photo above right, you can see the followers are held up high and clear of the camshaft lobes allowing uninterrupted work on the camshaft, cam bearings and timing gears.

3.0 Implementation Steps

1. Manufacture enough cam follower holding pins to hold each follower on the camshaft. This is dependent on the engine configuration.
2. Loosen off rocker gear, lift and pin all followers on the camshaft.
3. Complete camshaft and/or timing gear work.
4. Remove all pins and double check.
5. Tighten rocker gear and complete adjustment procedure.

4.0 Benefits

1. Significantly reduces the time to remove rocker gear, push rods and cam followers.
2. Reduces the contamination risks by not removing and cleaning disassembled components.
3. Safer working on camshaft and timing gear with all load removed from camshaft.
4. Eliminates the risk of mixing push rod and follower locations.
5. Faster job completion time.

5.0 Resources Required

1. 9/32 x 1 1/2" split pins
2. Tags that are durable and highly visible.

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6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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